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Integrated Flood Hazard and Settlement Vulnerability Assessment in Khagaria District, Bihar Using Geospatial Techniques and the AHP Method

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Abstract

The flood history of Khagaria district in Bihar has been catastrophic, with over 74% of the region susceptible to flooding. The significant floods that transpired in 1954, 1957, 1974, 1987, 2004, 2007, 2017, 2020, 2022, and 2024 profoundly affected individuals' lives, properties, and the district's agricultural sector. The current research seeks to employ a multicriteria decision-making framework within a Geographical Information System (GIS), integrating remotely sensed data, to delineate flood-prone areas in Khagaria district. The research employed an 11x11 decision matrix to assess the significance of each parameter in generating the final flood map via the Analytic Hierarchy Process (AHP). The parameters included are elevation, slope, rainfall, drainage density (DD), distance from river (DFR), topographic wetness index (TWI), stream power index (SPI), land use/land cover (LULC), normalised difference vegetation index (NDVI), lithology, and ruggedness index. The weights derived in the case are accepted, as evidenced by a consistency ratio (CR) of 0.0345 (less than 0.1). Khagaria district encompasses a total area of 1490.16 sq.km, of which 429.909 sq.km is designated as a high flood susceptible zone, and 399.502 sq.km as a very high flood vulnerable zone. The integrated 2024 flood inundation raster was utilized to calculate the impacted settlement areas by village (in hectares and percentage) and their inundation frequency, with corresponding maps prepared for the year 2024. The vulnerability map was validated by the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) methodology. The analysis produced an AUC value of 0.819 for the integrated inundation raster and 0.945 for the flood susceptibility zonation map, demonstrating a high degree of accuracy and a robust correlation with identified flood-prone areas, signifying a substantial association between the satellite flood data and the assessed flood-prone locations. The outcomes can substantially

improve local emergency preparedness efforts, infrastructure advancement, land-use planning, and flood risk mitigation strategies.

1 Introduction

Floods are among the most catastrophic and prevalent natural disasters globally (Mundhe et al. 2018). A flood hazard denotes the likelihood of a potentially damaging flood event of a particular magnitude occurring within a specified timeframe and geographic region (Brooks 2003). Severe flooding events frequently result in devastating consequences for the environment, economy, agriculture, infrastructure, and wildlife populations (Doocy et al. 2013). Flood annually inflicts billions of dollars in damages globally, leading to fatalities, destruction of property and infrastructure, and significant public health concerns (Sharma et al. 2001). Ighile et al. (2022) assert that it is the most lethal of all climate-related natural catastrophes, resulting in an estimated 80% mortality rate globally and incurring around $60 billion in yearly damages, with the devastation of agricultural land and existing infrastructure. Gangopadhyay et al. (2018) report that floods in India cause annual economic losses above USD 7500 million.

Vulnerability refers to the condition in which individuals, communities, assets, or systems are more susceptible to adverse impacts from risks due to environmental, social, economic, or physical factors (United Nations Office of Disaster Risk Reduction, 2009). The vulnerability of compromised ecological and socioeconomic systems, indicative of their susceptibility to hazardous events, dictates the magnitude of flood damage (Messner et al., 2005). Academics and policymakers have used indicator-based methodologies to assess vulnerability (Birkmann 2006). Fernandez et al. (2016) assert that vulnerability assessments and mapping provide comprehensive insights into the present circumstances and potential impacts of hazards on individuals, assets, and environments. While total eradication may be unattainable, precise identification of flood-prone regions can significantly mitigate the effects of flood hazards (Sahoo and Sreeja 2017).

A logical data structure-based indicator approach is employed in vulnerability assessments. The Analytical Hierarchy Process (AHP) is a recognised multicriteria decision-making technique that offers a systematic approach to addressing complex problems (Vijith & Dodge-Wan, 2019). This technique aims to estimate potential risk and reaction capabilities in hazard-prone areas (Chan et al. 2022). AHP is crucial for establishing the hierarchical framework of selected indicators, while GIS-based mapping visually represents the distribution of vulnerability by highlighting the most and least susceptible areas (Das and Pal 2020). Geographical decision-making necessitates weighting and ranking to integrate multiple factors for evaluating spatial vulnerability (Wang et al. 2011). Numerous factors, such as stakeholder engagement, empirical data, data accessibility, and study aims, affect the use of metrics for assessing flood-prone regions (De-Brito et al. 2018). The integration of AHP and GIS facilitates the assessment of flood hazards in a study area (Ibrahim et al. 2024). Flood susceptibility and inundation maps are essential for flood mitigation and risk management, since they offer accurate geographic data that aids in planning, risk assessment, and comprehension of landscape susceptibility (Prasad et al. 2022

A low-cost way for creating vulnerability maps is the combination of GIS with Hierarchical Multi-criteria Analysis (AHP). Kumar and Jha (2023) utilise ArcGIS 10.3 and the Shuttle Radar Topography Mission-Digital Elevation Model (SRTM-DEM) 30m DEM to create a flood risk map through the integration of AHP and GIS. Principal elements contributing to floods are height, slope, drainage capacity, precipitation, and land use. Thematic layers are assessed using the Analytic Hierarchy Process (AHP), incorporating consistency checks to guarantee reliability. The findings highlight the flood susceptibility of Purnia and Madhepura, facilitating informed decision-making. Bapalu et al. (2008) utilized the multi-parametric AHP method to evaluate flood risk in the Kosi River basin, employing GIS and hydrological analysis to generate risk maps. They further delineated topography, population density, land cover, and geomorphology, integrating a flood risk score corroborated by hydrological inundation data. This method was identified as a cost-efficient strategy for formulating flood mitigation plans.

Approximately 74% of Khagaria's total area is classified as flood-prone, rendering it highly susceptible to flooding. Significant floods occurred in 1954, 1957, 1974, 1987, 2004, 2007, 2014, 2017, 2020, 2023, and 2024, causing extensive damage to agricultural land, property, and human lives (Flood Hazard Atlas Bihar 2020). Extensive land regions, including cultivated crops, are inundated annually during catastrophic floods, leading to substantial agricultural losses of approximately one hundred thousand hectares and numerous fatalities (Kumar 2019). There is a lack of research on flood risk assessment in the Khagaria district of Bihar, a notable flood-prone area in India. A multitude of scholars has investigated the flooding phenomenon and projections in the North Bihar region through the periodic utilisation of multi-temporal satellite datasets and GIS technologies. However, there is inadequate data on current flooding trends to determine flood risk zones and mitigation strategies in Khagaria (Tripathi et al. 2022). Therefore, the present study aims to identify flood-prone areas in the Khagaria district and propose appropriate flood mitigation strategies by employing a multicriteria decision-making approach inside a GIS framework, utilizing remotely sensed data.

2 Study Area

Khagaria district consists of two subdivisions: Khagaria and Gogari. The district is situated at roughly 25°30' N latitude and 86°29' E longitude (or 25.5°N, 86.48°E) and encompasses an area of 1,485 square kilometres.

The alluvial plains have a saucer-shaped depression that remains marshy for most of the year and is flooded during the monsoon as a result of river overflows. The district is traversed by significant rivers, including the Burhi Gandak, Bagmati, Kamla, Ganga, and Ghaghri, a key tributary of the Koshi river system. Khagaria's climate oscillates between the arid, sweltering heat of the northern plains and the wet, moisture-rich environment of the southern Bengal basin. The district lacks notable hills or mineral resources, with agriculture as the primary economic activity (Kumar 2019). Khagaria is highly susceptible to flood disasters due to its low-lying topography, tropical climate, and the abundance of rivers. Figure 1 illustrates the research area map created via a False Colour Composite (FCC) of Sentinel-2 data obtained on 27 December 2024. The picture emphasises plant and land cover characteristics, offering a more distinct spatial context of Khagaria district. Khagaria is a low-lying area that is traversed by important rivers, such as the Burhi Gandak, Bagmati, Kamla, Ganga, and Ghaghri, a major tributary of the Koshi river system. The district's flood is caused by the heavy rainfall in the Nepal region and northern Bihar, which overflows these rivers with water.

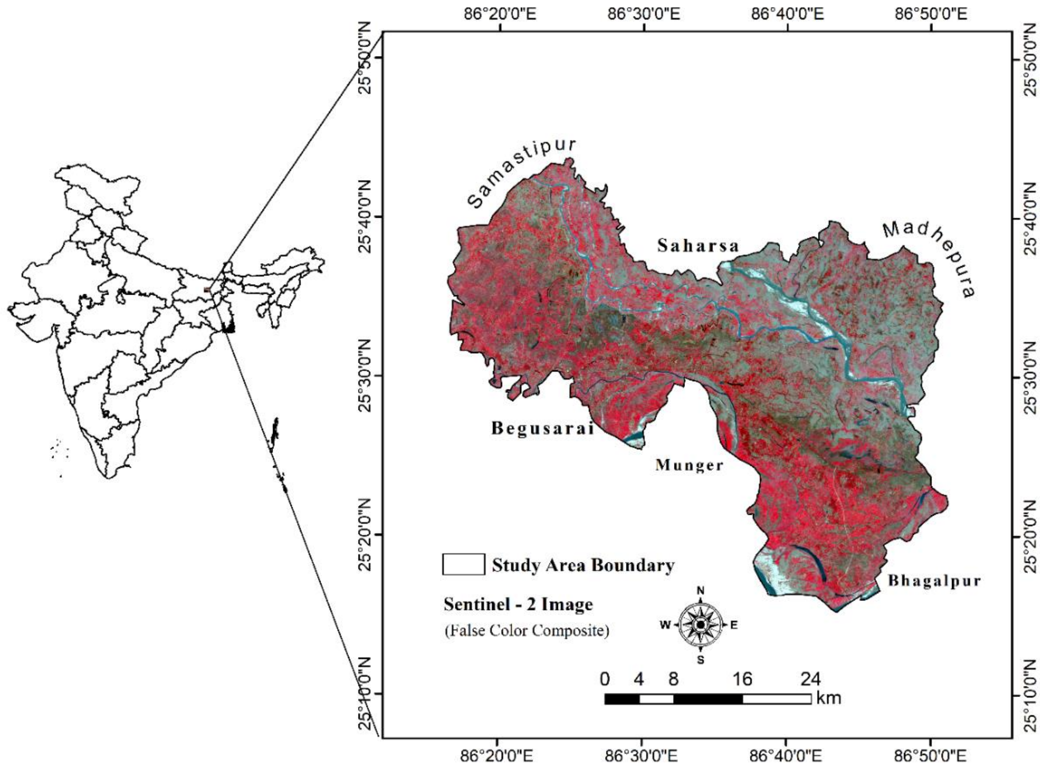


Figure 1 Study area.

3 Material

Multiple datasets (as illustrated in Table 1) were collected for the current study to develop the desired thematic flood layers. The precipitation map was generated with Climate Research Unit (CRU) data, obtained from https://crudata.uea.ac.uk/cru/data/hrg/. The SRTM data was obtained free of charge from the USGS website (https://earthexplorer.usgs.gov/). The elevation, slope, Topographic Wetness Index (TWI), ruggedness index, river pattern of the watershed, and water basins of the research area were derived from the SRTM DEM data and a remotely sensed image obtained from the Sentinel-2A sensor (10 m spatial resolution) in May 2024, which was utilized for visual interpretation of land use and land cover via ArcGIS software. Sentinel-1 VV/VH polarisation data (10 m spatial resolution) was utilized to create a date-specific flood inundation layer, which facilitated the validation of the flood susceptibility map developed using the AHP technique. The Sentinel datasets were acquired via the Copernicus Open Access Hub website (https://www.copernicus.eu/en/access-data/conventional-data-access-hubs). Topographic sheets (Topo-Sheets) Nos. 72K/11, 72K/10, 72K/6, 72K/7, 72K/14, and 72K/15 (2010) of Khagaria district, at a scale of 1:50,000, were obtained from the Survey of India (SOI) website (https://www.surveyofindia.gov.in/) for comparison with the LULC map. Population data from 2011 was obtained from the Census of India (2011) website (https://www.census2011.co.in/). Lithological data was acquired from the Geological Survey of India (GSI) portal (https://www.gsi.gov.in/webcenter/portal/OCBIS) to create the lithology map.

Table1 Data Types.

SNO.	DATA TYPES	SOURCES
1.	Rainfall Data	IMD, Pune (0.25X0.25 resolution) https://www.imdpune.gov.in
2.	SRTM-DEM (30m resolution)	USGS Earth Explorer website (https://earthexplorer.usgs.gov/).
3.	OSM Toposheet No.- 72K/11, 72K/10, 72K/6, 72K/7, 72K/14, 72K/15 (2010)	Survey of India (SOI), https://www.surveyofindia.gov.in/
4.	Sentinel-1 Data (10 m spatial resolution)	Copernicus Open Access Hub https://www.copernicus.eu/en/access-data/conventional-data-access-hubs
5.	Sentinel-2 Data (10 m spatial resolution)	Copernicus Open Access Hub https://www.copernicus.eu/en/access-data/conventional-data-access-hubs
6.	Lithology Data	Geological Survey of India (GSI), https://www.gsi.gov.in/webcenter/portal/OCBIS

4 Methods

Prior to analysis, all preliminary procedures were completed, encompassing downloading, extracting, georeferencing, formatting, and resampling the digital data of the variables (Hagos et al. 2022). The study was conducted in accordance with this methodological framework, as shown in Figure (2) ensuring that all spatial data were geometrically and radiometrically adjusted and rectified to a standardised projection coordinate system. The three primary phases of the methodological framework—data collection, data processing, consistency verification, and vulnerability map creation—are illustrated in Figure 2.

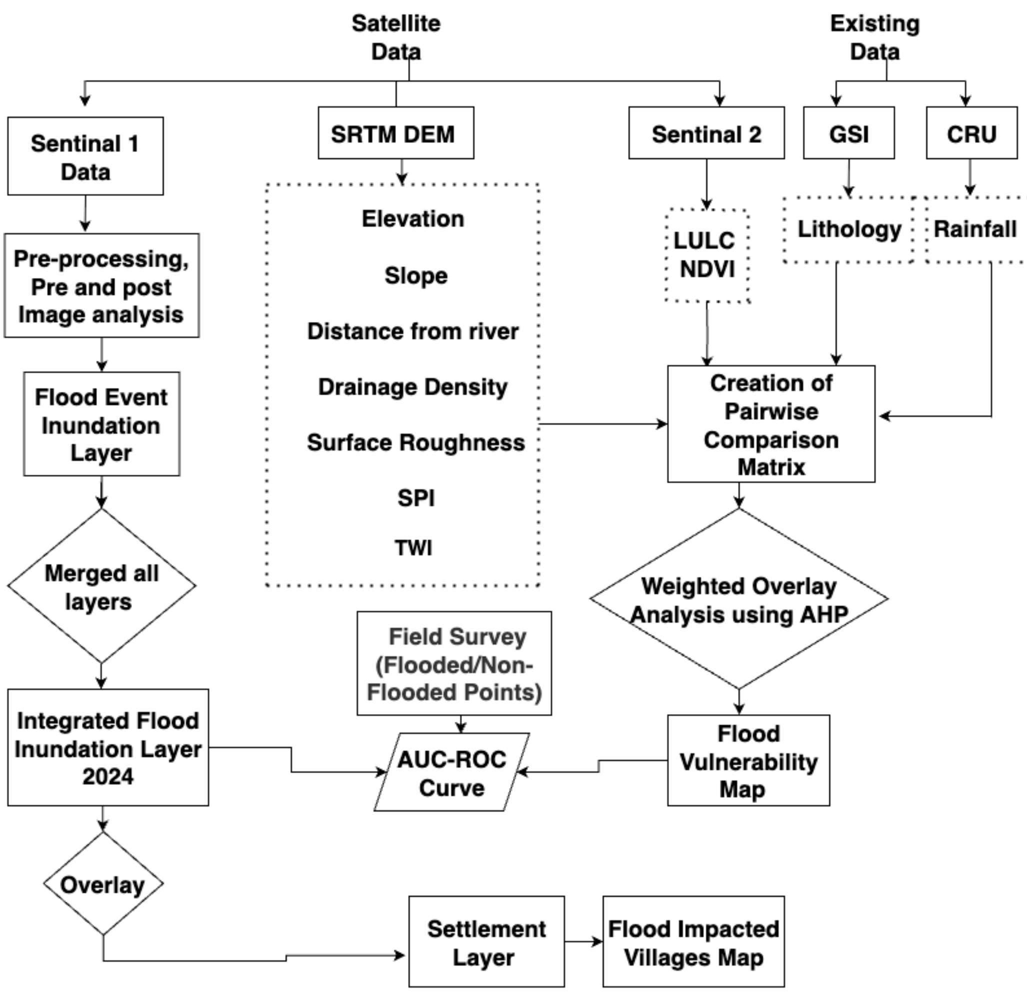


Figure 2 Methodology flowchart.

4.1 Decision criteria for identification of flood vulnerable zone and selection of flood-influencing parameter

The Analytic Hierarchy Process (AHP), introduced by Saaty in 1980, is a semi-quantitative, multicriteria decision-making methodology that involves determining choices through systematic pairwise comparisons of elements (Das 2019). The AHP methodology facilitates priority establishment and enhances decision-making by methodically assessing and contrasting various criteria. Flood development in the study area is influenced by a combination of natural and anthropogenic factors (Dandapat & Panda 2017). Table 2 delineates the significance of each affecting parameter. This study offers a thorough approach for assessing flood risk that uses the Analytical Hierarchy Process (AHP) to combine physical hazard indicators with socioeconomic vulnerability elements. The Flood Hazard Zonation (FHZ) was created using biophysical factors like elevation, slope, distance from rivers, rainfall, topographic wetness index (TWI), modified normalised difference water index (MNDWI), and normalised difference vegetation index (NDVI), while the Flood Vulnerability Zonation (FVZ) was based on demographics, land use/land cover (LULC), accessibility to essential services, and illiteracy rates. Important criteria (elevation, slope, drainage density, distance from stream, rainfall, population density, land use, land cover, soil bulk density, flood frequency) were weighed using the Analytic Hierarchy Process (AHP). The Gandak River watershed saw about 118 floods over the course of eight years, according to the flood frequency map (Bhat et al., 2025). North Bihar's Lower Kosi Basin (LKB), which is impacted by upstream hydrology, is extremely vulnerable to flooding. Eleven flood conditioning parameters (precipitation, elevation, slope, drainage density, distance from the river, ruggedness index, topographic wetness index, stream power index, curvature, normalised difference vegetation index, land use, and land cover) obtained from the satellite data were integrated using a weighted linear summation model to create a flood susceptibility index (Shivhare, 2024). (Kumar and Jha, 2023). The selection of the study region, the identification of the flood-causing causes and the gathering of necessary data, the creation of the desired thematic layers, and their integration to create the flood risk map comprise the study process. The theme layers—slope, district's distance to an active stream, highest elevation, drainage density, rainfall, population density, and land use-land cover—have been chosen for AHP analysis for flood vulnerability map.

Table 2 Weights of each parameter and sub-parameters.

Parameters	AHP Weights	Class Range	Flood Level	Rank
Elevation	21.8	0-30m	Very High	5
		31-37m	High	4
		38-42m	Moderate	3
		43-50m	Low	2
		>50m	Very Low	1
Slope	20.3	0-1.1	Very High	5
		1.2-2.3	High	4
		2.4-3.7	Moderate	3
		3.8-6.3	Low	2
		6.4-36	Very Low	1
DFR	15.8	0-500m	Very High	5
		501-1000m	High	4
		1001-2000	Moderate	3
		2001-3000	Low	2
		above 3001	Very Low	1
Rainfall	13.8	938.53-982.08	Very Low	1
		982.09-1010.85	Low	2
		1010.86-1041.18	Moderate	3
		1041.19-1076.17	High	4
		1076.18-1136.83	Very High	5
DD	8.3	0-1.7	Very Low	1

Parameters	AHP Weights	Class Range	Flood Level	Rank
		1.8-3.3	Low	2
		3.4-5.3	Moderate	3
		5.4-7.7	High	4
		7.8-14	Very High	5
SPI	5.3	(-5.1- 0.33)	Very High	5
		(-0.32- -0.13)	High	4
		(-0.12-0.0056)	Moderate	3
		(0.0057-0.41)	Low	2
		(0.42 - 3.4)	Very Low	1
LULC	4.7	Vegetation	Very Low	1
		Cropland	Low	2
		Barren Land	Moderate	3
		Built up	High	4
		Waterbody	Very High	5
Ruggedness Index	3.4	0.11-0.3	Very High	5
		0.31-0.43	High	4
		0.44-0.56	Moderate	3
		0.57-0.69	Low	2
		0.7-0.89	Very Low	1
TWI	2.7	(-7.8 - - 3.5)	Very Low	1
		(-3.4- -1.2)	Low	2
		(-1.1- 1)	Moderate	3
		(1.1-3.8)	High	4
		(3.9-12.2)	Very High	5
Lithology	2.2	Holocene	Very High	5
		Meghalayan	Moderate	3
NDVI	1.7	< 0.25	Very High	5
		0.26-0.4	High	4
		0.41-0.55	Moderate	3
		0.56-0.65	Very Low	2
		> 0.65	Low	1

Elevation

The height above sea level is referred to as elevation. Elevation is often articulated in feet or meters (Singh 2011). Elevation is a primary determinant of flooding; typically, the probability of flooding escalates as elevation diminishes (Choubin et al. 2019). Elevation layer generated from the SRTM (DEM) with a precision of 30 meters. The altitude in the Khagaria district study area varies from 0 to 90 meters. The elevation layer was reclassified into five categories: 0–30 m (very low elevation), 31–37 m (low elevation), 38–42 m (moderate elevation), 43–50 m (high elevation), and >50 m (very high elevation). In the weighted analysis, lower elevation classes received higher weights, whilst higher elevation classes were allotted lesser weights, indicating their proportional impact on flood susceptibility. The elevation map of Khagaria is represented in Figure 3.

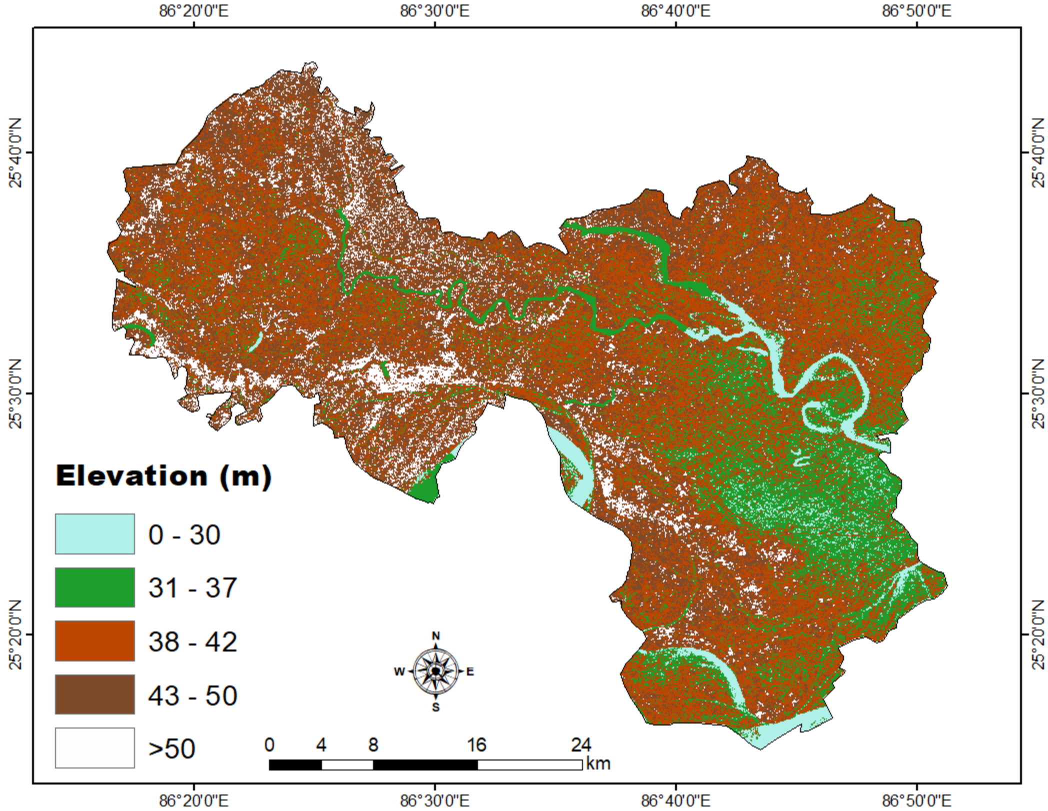


Figure 3 Elevation map.

Slope

The elevation differential between two points per unit of distance is termed the slope. The slope is a pivotal element in hydrology, as it substantially affects surface runoff and, subsequently, the incidence of floods (Meraj et al. 2015). To generate a slope map, we must initially import a 30 m resolution SRTM Digital Elevation Model (DEM) into ArcGIS, which is subsequently transformed into a slope map utilizing the slope function within the Spatial Analyst tool in ArcGIS. The slope map was categorised into five classifications according to degrees of inclination: very mild (0–1.1°), gentle (1.2–2.3°), moderately steep (2.4–3.7°), steep (3.8–6.3°), and extremely steep (6.4–36°). Slope inclination inversely impacts infiltration rates; as the gradient increases, the capacity for infiltration diminishes. A mild slope is given significant weight, whereas a steep slope is awarded minimal weight. The slope map of Khagaria is represented in Figure 4.

The slope can be determined using Equation (1):

Slope Calculation Formula:

$$\text{Slope in degrees} = \arctan(\Delta Z/\Delta X) \times 180/\pi \quad (1)$$

Where:

ΔZ = change in elevation (vertical distance),

ΔX = horizontal distance (Esri 2020).

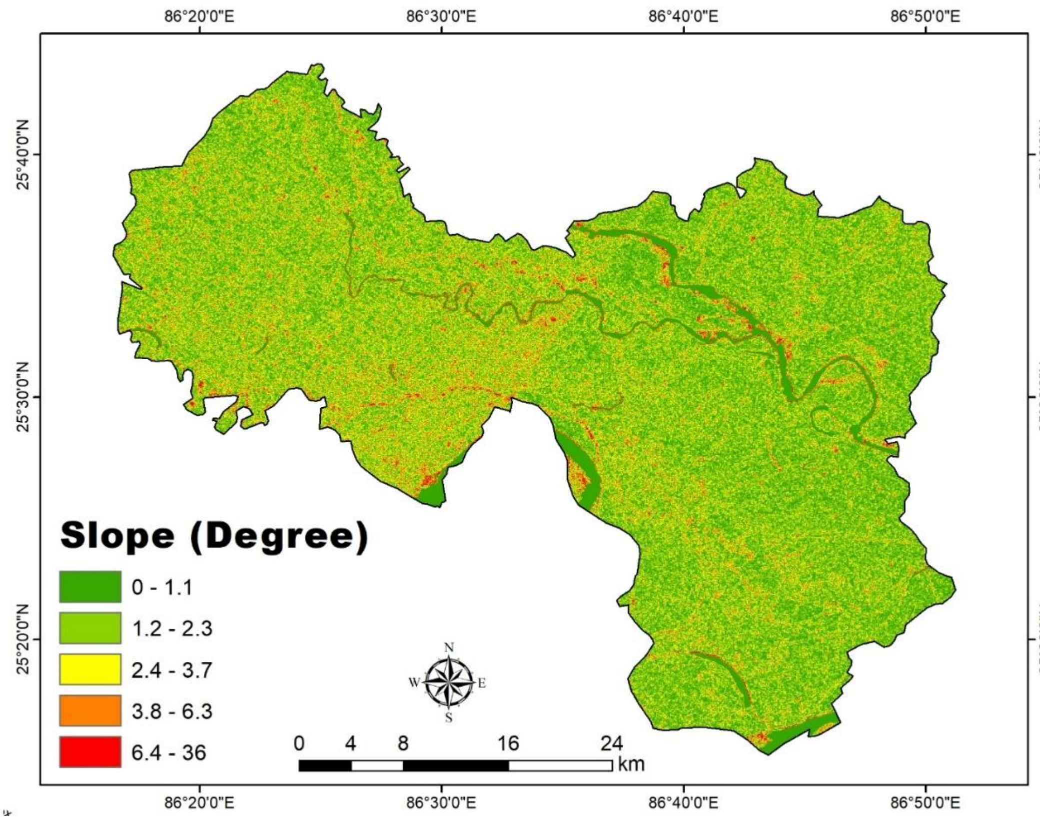


Figure 4 Slope map.

Distance from River

The proximity to rivers is essential when determining the flood risk zone (Chapi et al. 2017). The influence of flood threats is greater in proximity to the river network and diminishes with increasing distance from the river (Bora et al. 2022). Utilizing SRTM (DEM 30m) and the Euclidean distance tool in ArcGIS, the DFR map was generated and categorised into five groups based on proximity to the river: 0-500 m indicating a very high category, 501-1000 m indicating a high category, 1001-2000 m indicating a moderate category, 2001-3000 m indicating a low category, and above 3001 m indicating a very low category. The distance from river map of Khagaria is represented in Figure 5.

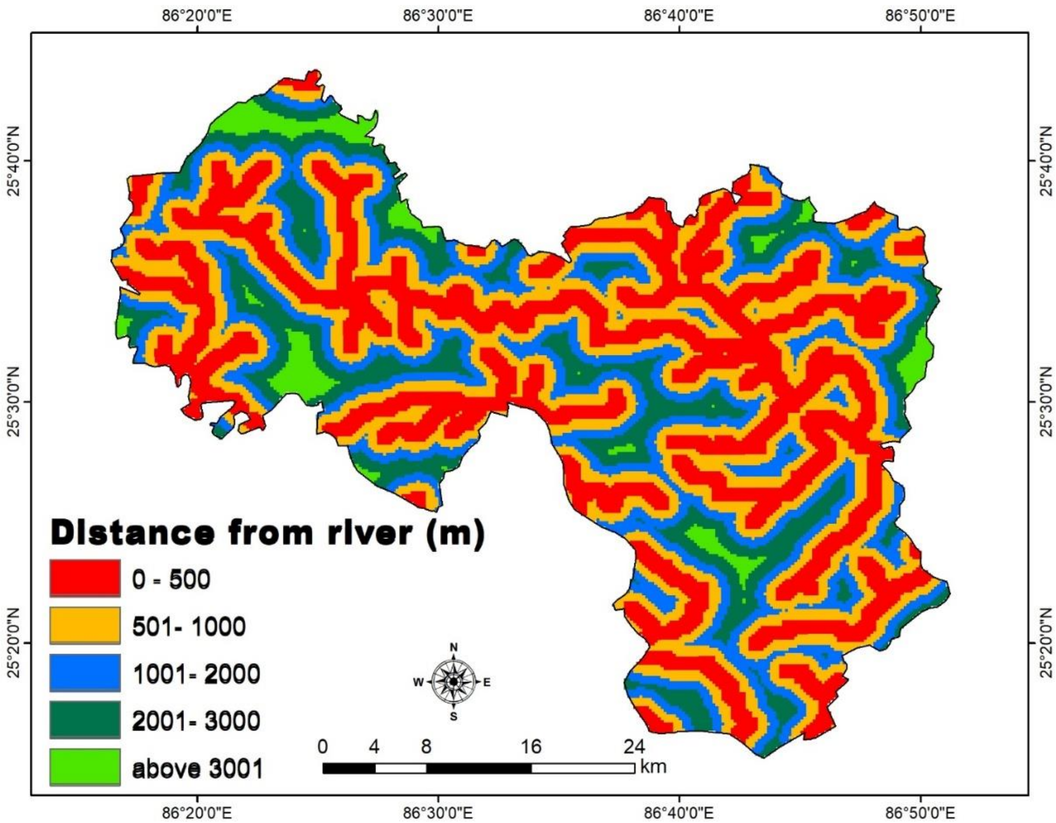


Figure 5 Distance from the river map.

Rainfall

Precipitation is a vital factor in flood susceptibility assessment, as it directly affects river inundation, which subsequently supports the local flood web (Osman & Das 2023). Enhanced precipitation depth generally results in substantial flooding, whereas reduced precipitation depth causes negligible flooding. This work developed a rainfall map utilizing data from the IMD, Pune portal, which was interpolated via the kriging interpolation method in ArcGIS to provide a continuous spatial representation. The interpolated rainfall data at a 30m resolution were categorised into five classifications: very low rainfall (938.53–982.08 mm), low rainfall (982.09–1010.85 mm), moderate rainfall (1010.86–1041.18 mm), high rainfall (1041.19–1076.17 mm), and very high rainfall (1076.18–1136.83 mm). The classes were utilized to evaluate the spatial variability of precipitation and its impact on flood susceptibility, assigning greater weights to higher rainfall classes and lesser weights to lower rainfall classes. The rainfall map of Khagaria is represented in Figure 6.

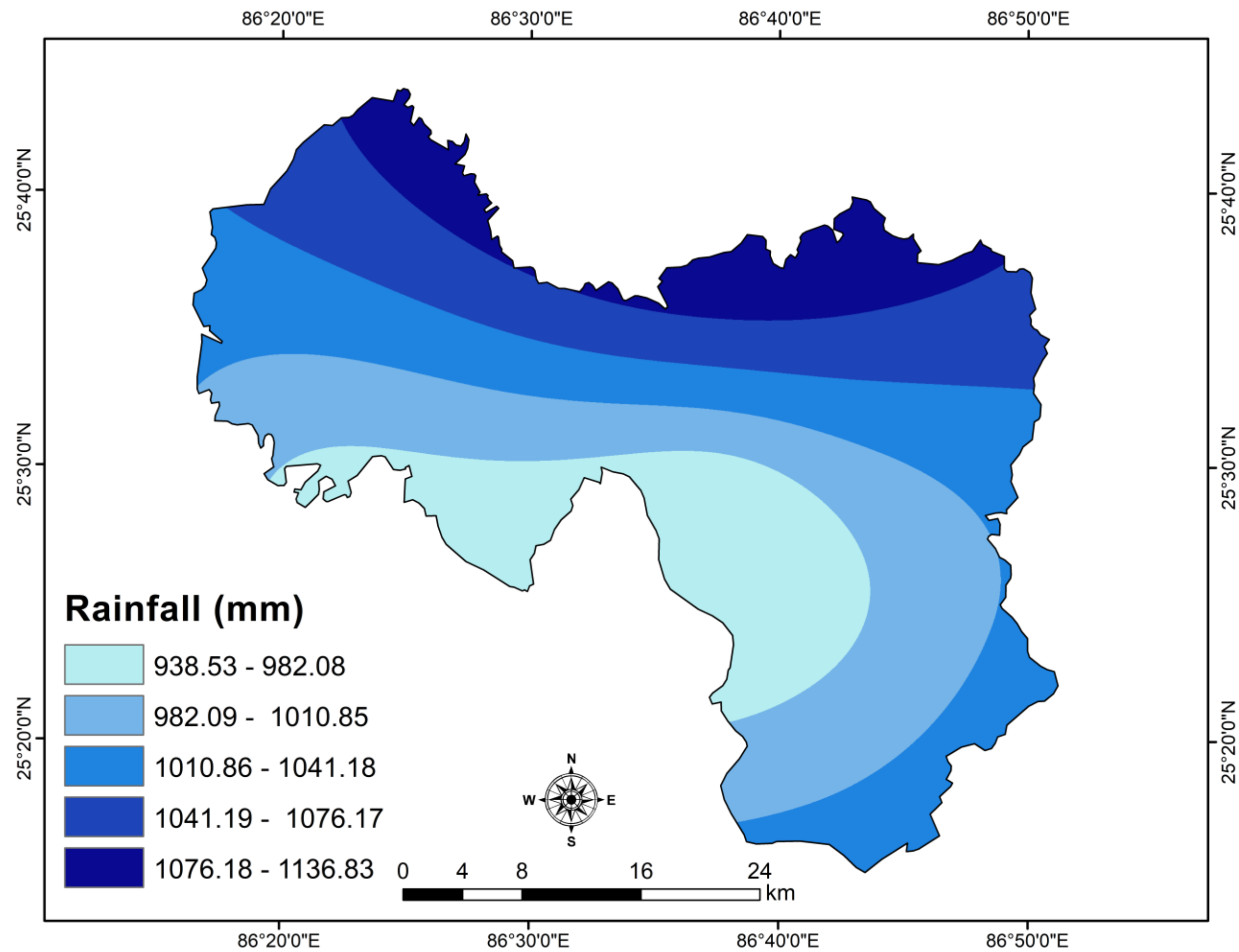


Figure 6 Rainfall map.

Drainage density

R.E. Horton coined the term "drainage density" in 1932 to denote the total length of streams per unit area. It is a crucial hydrological metric, as elevated DD signifies increased flood susceptibility. The probability of flooding is directly related to drainage density; an extensive drainage network leads to heightened surface runoff, consequently increasing the risk of flooding (Geetha et al. 2024). The DD map was subsequently generated utilizing the Line Density tool in ArcGIS. The drainage density map of Khagaria is represented in Figure 7.

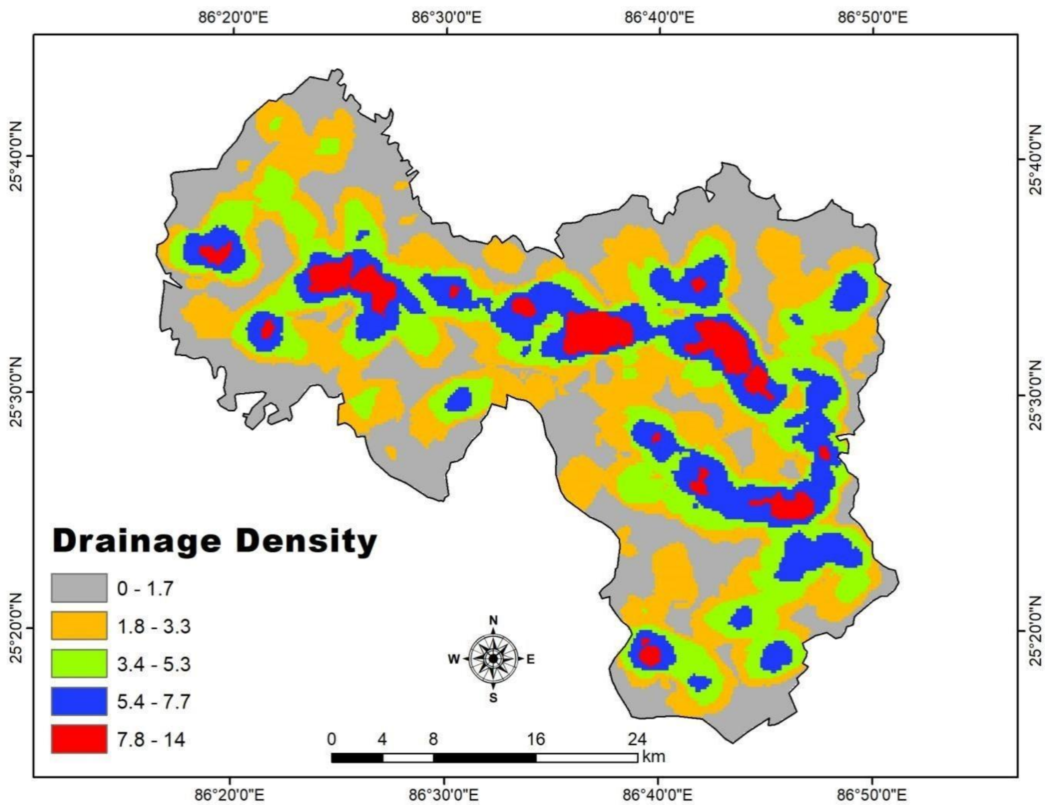


Figure 7 Drainage density map.

The DD was calculated using the following Equation:

$$D = \sum L / A \quad \dots \qquad (2)$$

Where:

ΣL = total length of the hydrographic network (km), and

A = hydrographic basin area (km^2) (Moeini et al. 2015).

The DD has been classified into five categories, as flood susceptibility escalates with DD. Regions characterised by very low (0–1.7 km/km^2) and low (1.8–3.3 km/km^2) drainage densities demonstrate a low to moderate flood risk attributable to increased infiltration and diminished runoff. Conversely, moderate (3.4–5.3 km/km^2) to very high (7.8–14 km/km^2) classifications signify elevated flood susceptibility, as dense stream networks result in augmented surface runoff and expedited water accumulation.

Stream Power Index

Cao et al. (2016) state that the SPI quantifies a discharge's erosive capacity relative to the specific area of a watershed. It measures the intensity of water flow at a designated location within a watershed. The intensity of the flow escalates with the SPI value (Bora et al. 2022). Negative SPI values generally arise from concave topography and regions where runoff quickly converges; consequently, the more negative the value, the higher the flood risk. Positive SPI values typically signify flat or convex topography, leading to reduced water flow and diminished flood risk. The SPI map of Khagaria is represented in Figure 8.

SPI is denoted as shown in Equation (3):

$$SPI = AS \times \tan(\beta) \quad (3)$$

Where:

AS = specific catchment area (m²/m) — the upslope contributing area per unit width orthogonal to the flow direction, and
tan(β) = Slope in radians — the local slope gradient.

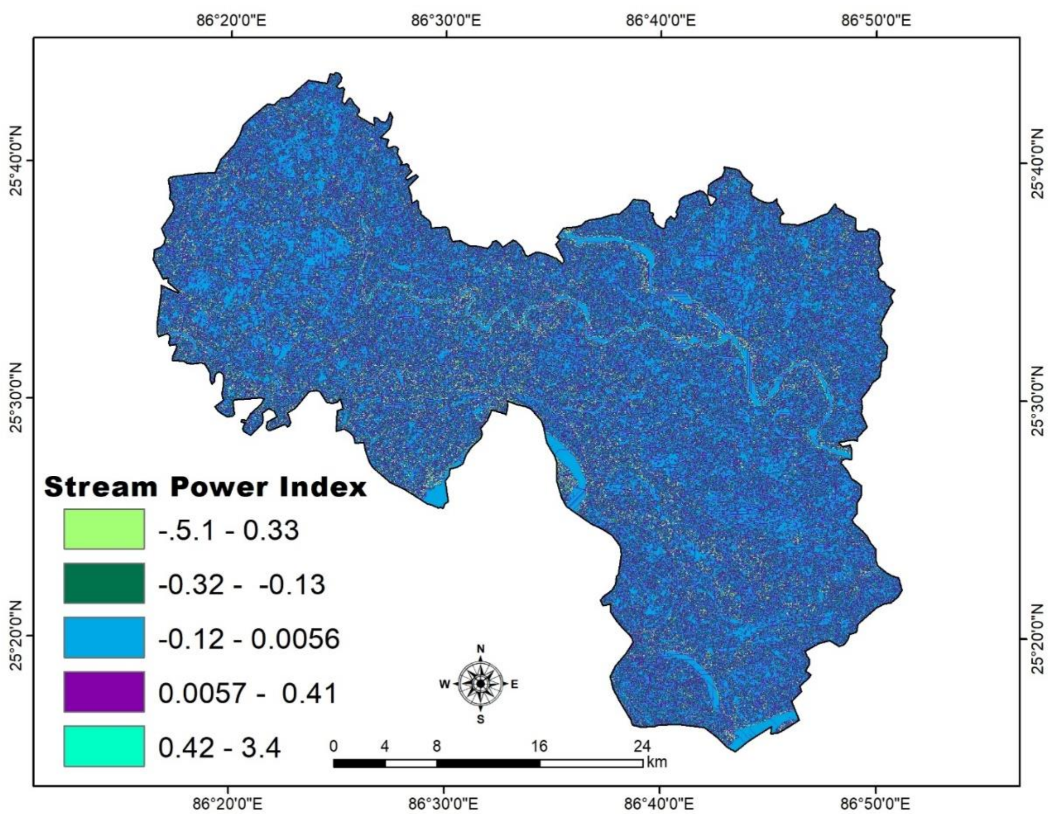


Figure 8 SPI map.

SPI prepared from SRTM-DEM (30m resolution). It has been categorized into five classes: (-5.1- -0.33) represents a very high SPI, (-0.32- -0.13) represents a high SPI, (-0.12- 0.0056) represents a moderate SPI index, (0.0057- 0.41) represents a low SPI and (0.42- 3.4) represents a very low SPI.

LULC

Vulnerability mapping is crucial for identifying activities in a location and the various types of land use and land cover (LULC) impacted by recurrent flooding (Fernandez & Lutz 2010). Land Use and Land Cover (LULC) was generated utilizing Sentinel-2 data, which underwent supervised classification into five categories: water bodies, vegetation, built-up areas, cropland, and bare ground. Land cover classifications are assigned weights based on their influence on surface runoff and infiltration. The LULC map of Khagaria is represented in Figure 9.

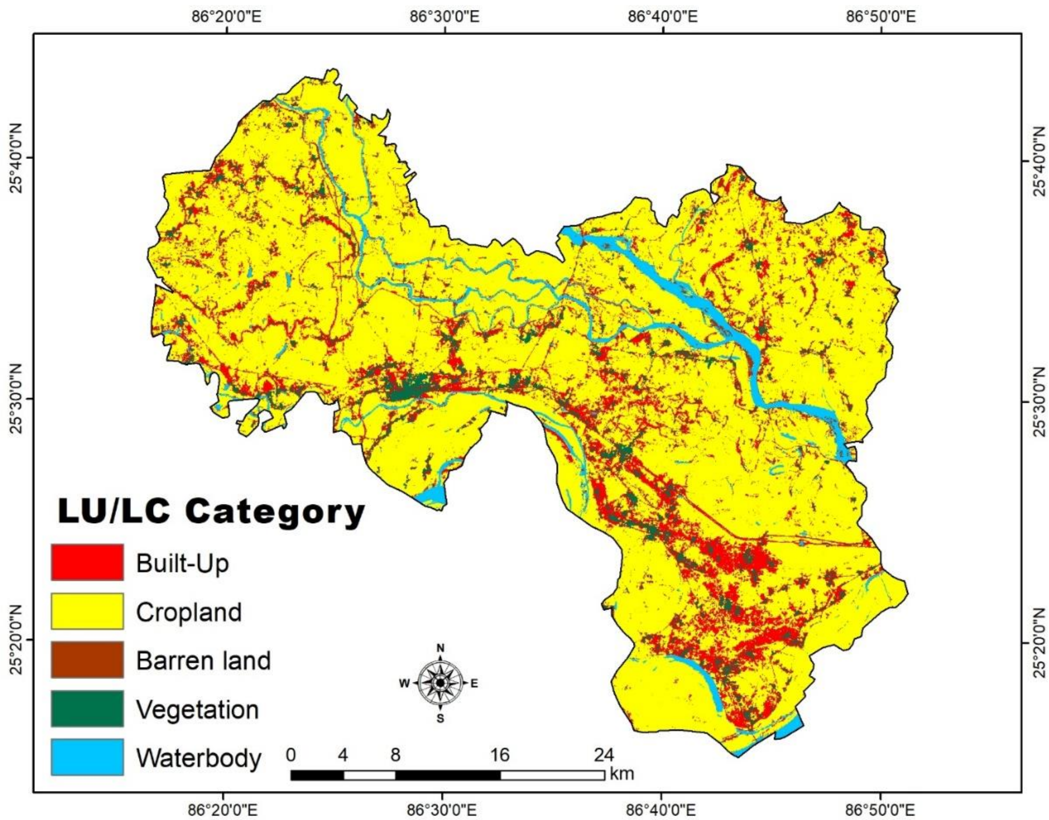


Figure 9 LULC map.

The LULC map of the study area is generated by ArcGIS and multi-spectral Sentinel-2 data and has been classified from the supervised classification method implemented with spectral signatures derived from the training samples (Mann & Gupta 2023). LULC was prepared from Sentinel- 2 data classified in five classes—water bodies, vegetation, built-up areas, agriculture, and barren land.

Ruggedness index

The ruggedness index (Strahler 1950) signifies the extent of land surface instability. Topography and drainage are taken into account when determining the ruggedness index (Shankar & Dharanirajan 2014). Precipitation patterns, land use, and soil type collectively influence the extent to which roughness affects flooding risk (Morrison et al. 2017). It frequently combines length and gradient steepness. When the slopes of the basin are both elongated and steep, they yield exceptionally high ruggedness values. The ruggedness index map of Khagaria is represented in Figure 10. Strahler (1968) posits that the roughness index is derived from the product of drainage density (Dd) and maximum basin relief (R), as illustrated in Equation (4):

$$Rn = R \; x \; Dd \tag{4}$$

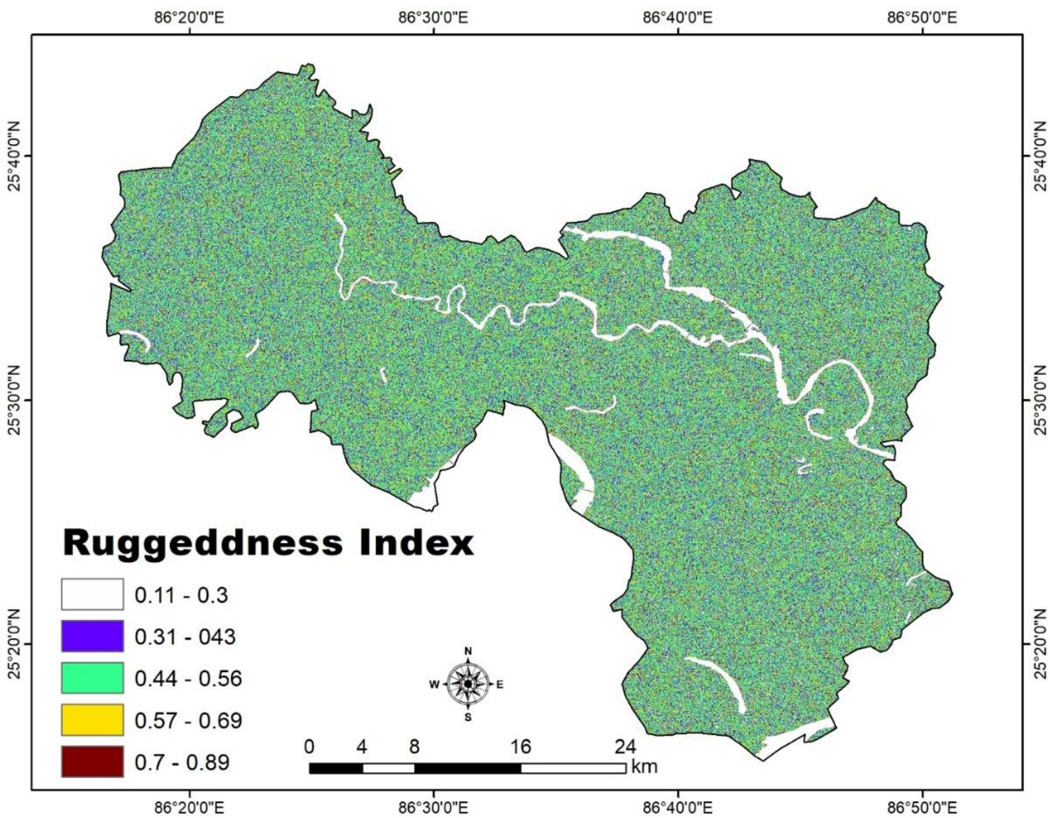


Figure 10 Ruggedness Index map.

The ruggedness index map has been created using focal statistics and the raster calculator in ArcGIS. It has been categorized into 5 classes as- (0.11- 0.3) represents a very high ruggedness index, (0.31- 0.43) represents a high ruggedness index, (0.44- 0.56) represents a moderate ruggedness index, (0.57- 0.69) represents low ruggedness index, (0.7- 0.89) represents very low ruggedness index.

TWI

The TWI reflects the soil moisture retained as a result of inadequate drainage and various other factors. TWI is a widely utilized metric for identifying and measuring regions that are water-saturated (Kaya & Derin 2023). The TWI map of Khagaria is represented in Figure 11.

The TWI is mathematically defined in Equation (5) as follows:

$$W = In\ \alpha / tan\ \beta\ \ldots \qquad (5)$$

The slope angle at that position is represented by β, and the W value is the wettability index, represented by α, which measures the volume of water gathered from the top Slope at each contour unit.

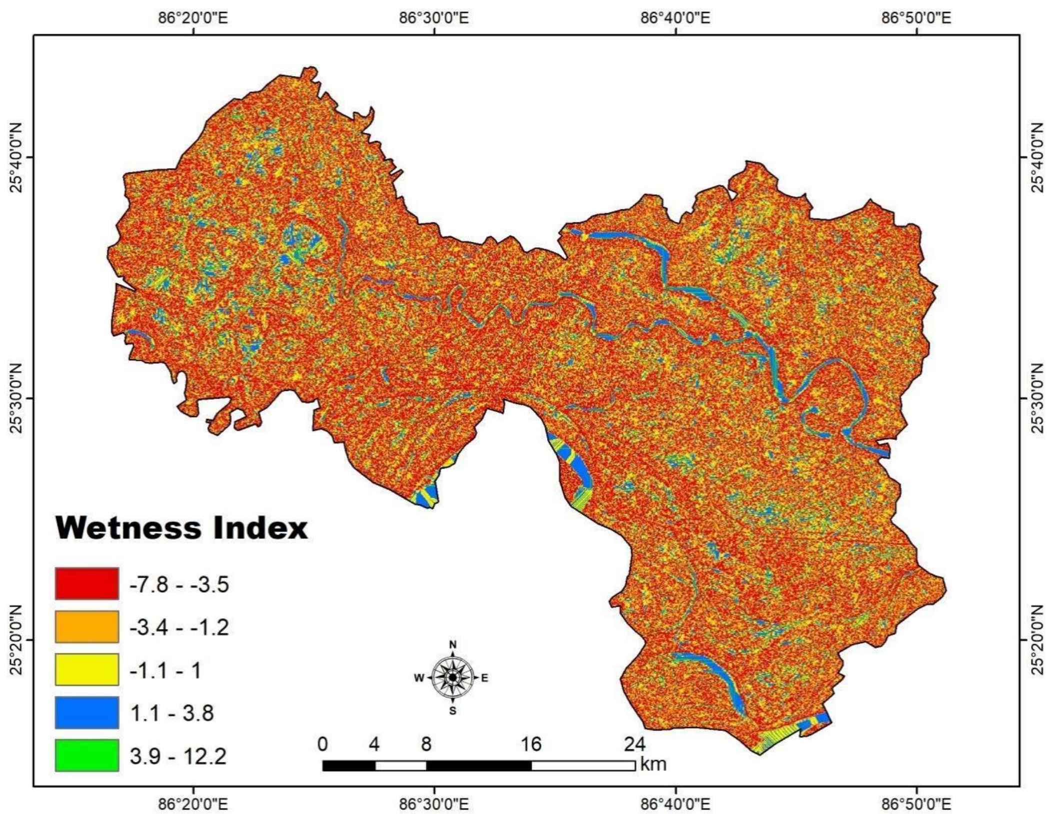


Figure 11 TWI map.

High soil moisture levels will result in a greater proportion of precipitation converting to surface runoff, thereby elevating flood susceptibility. The TWI is classified into five categories: (-7.8 to -3.5) indicates a very low wetness index, (-3.4 to -1.2) signifies a low wetness index, (-1.1 to 1) denotes a moderate wetness index, (1.1 to 3.8) reflects a high wetness index, and (3.9 to 12.2) represents a very high wetness index. The TWI was generated utilizing the raster calculator function within the ArcToolbox of ArcGIS software.

Lithology

Floods are affected by geological formations and various rock attributes, such as porosity, permeability, and the alterations in the voids present in rock joints and fissures. Consequently, elevated flooding will occur due to rocks with high resistance exhibiting low infiltration capacity (Shahabi et al. 2021). This study utilized lithology data obtained from the Geological Survey of India, which was subsequently converted into a 30m raster format. Lithological units are categorised into two classifications: Holocene formations, given significant importance, are deemed to significantly enhance flood susceptibility owing to their low infiltration capacity. Conversely, Meghalayan formations, designated a moderate weight, indicate moderate flood susceptibility. The lithology map of Khagaria is represented in Figure 12.

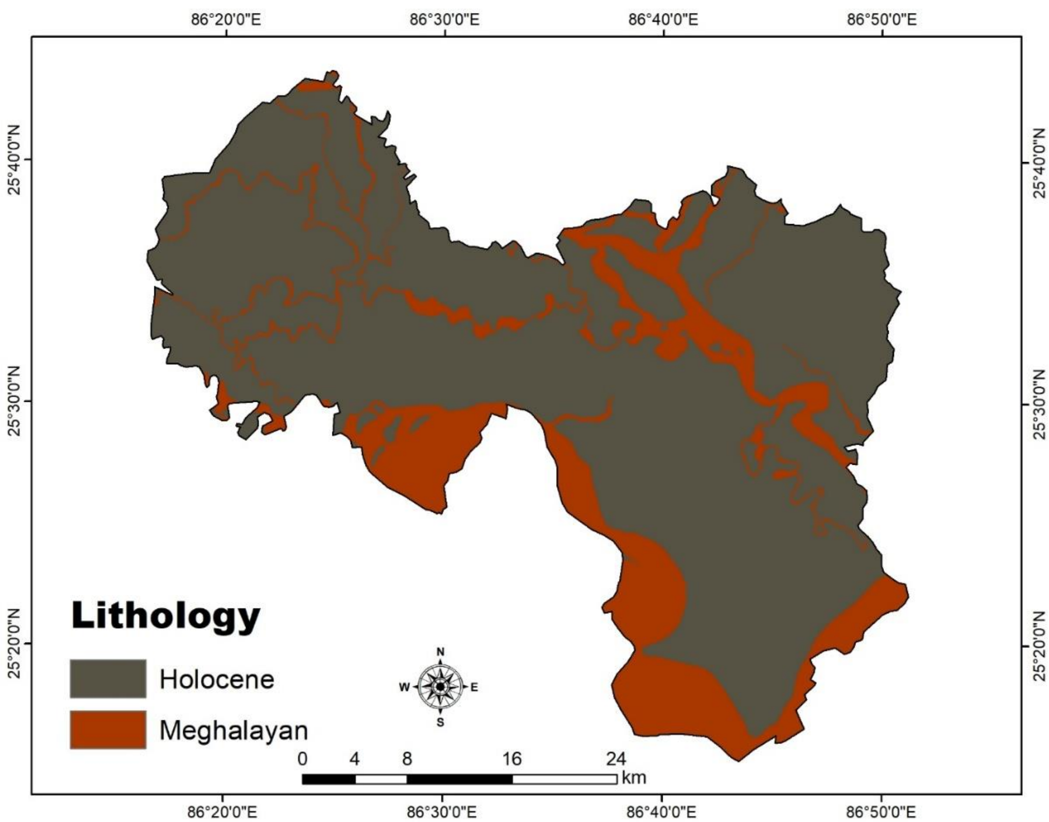


Figure 12 Lithology map.

NDVI

According to Sidle and Ochiai (2006), the NDVI is crucial for immobilizing massive amounts of water and enhancing the lithological mass's soil cohesiveness and shear resistance. NDVI represents vegetation health and density. It ranges from -1 to 1. Values closer to 1 indicate healthy vegetation. The NDVI map of Khagaria is represented in Figure 13. It is calculated using red and near- infrared bands using Equation (6):

$$NDVI = (NIR-R)/(NIR+R) \tag{6}$$

Where:

NIR = Near infrared reflectance, and

R = Reflectance.

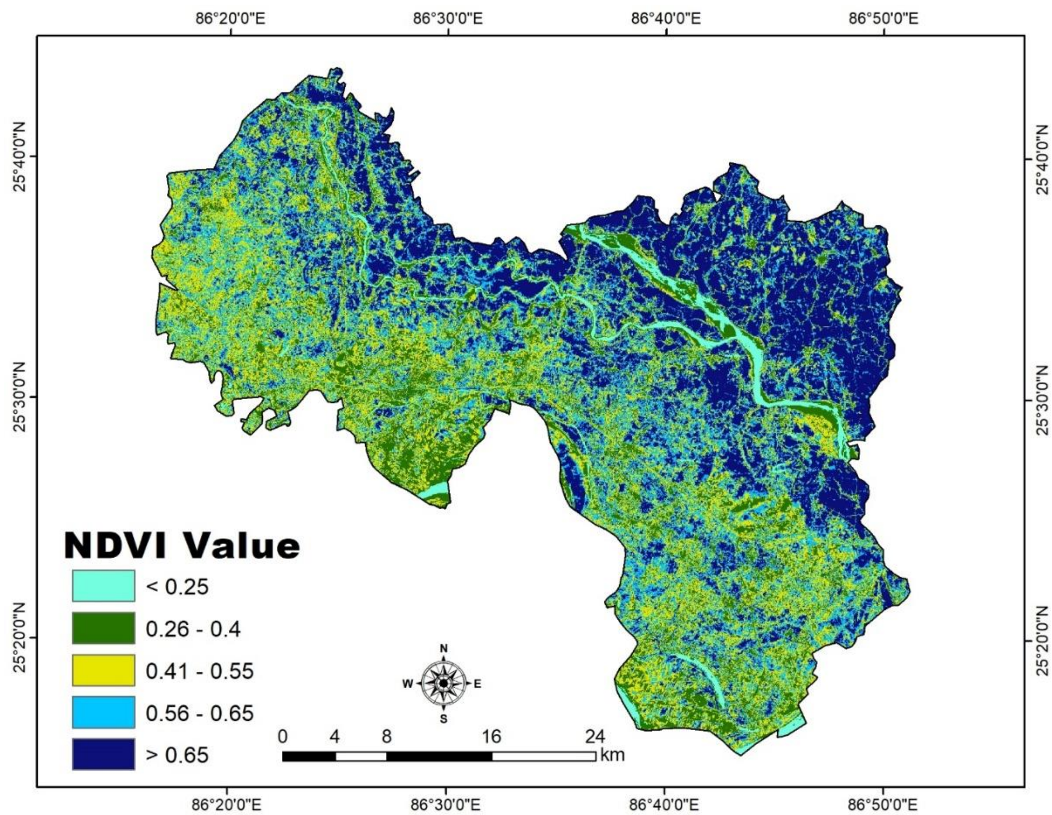


Figure 13 NDVI map.

In the healthy forest region, the NDVI value is elevated, and water infiltration into the soil is also substantial. Low values are given significant weight. The NDVI map was generated utilizing a Sentinel-2 image. The classification consists of five categories: (< 0.25) indicates very high NDVI, (0.26-0.4) indicates high NDVI, (0.41-0.55) indicates moderate NDVI, (0.56-0.65) indicates low NDVI, and (> 0.65) represents very low NDVI.

4.2 Reclassifying datasets

Reclassification criteria serve multiple functions, including updating values based on new information, categorising entries, reclassifying values according to a standardised scale, designating certain values as "No Data," or assigning a value to "No Data" cells. The reclassification functions employ various techniques to convert cell values into alternative values (Mundhe et al. 2018). The present study project identified 11 significant factors to delineate the flood-prone area. Subsequently, reorganise the raster layers utilizing the ranking method within the range of 1 to 5. In this context, elevated figures signify increased flood susceptibility. Ultimately, these layers were amalgamated or superimposed utilizing the map algebraic expression and the "Raster Calculator" function of the ArcGIS Spatial Analyst tool. The flood layer map is ultimately generated by consolidating all the aforementioned layers. (Mundhe et al. 2017).

4.3 Pairwise comparison matrix

The Weighted Overlay tool in the Spatial Analyst tool extension integrates the layers in the ArcGIS software to generate maps of flood susceptibility (Saaty 1977). The impact of each thematic layer and the associated scale values, derived from the weights and scores determined by the AHP method is implemented (Kumar & Jha. 2023). The simple 1-9 intensity scale was employed to develop multiple paired criteria comparison matrices derived from expert opinions and prior research (Zenande et al. 2024), as illustrated in Table 3. The pairwise comparison matrix has been constructed for all the 11 parameters and weights have been assigned to each parameter priority wise as shown in Figure 14.

Table 3 Fundamental 9-Point Intensity Importance Scale.

Intensity of Importance - Definition	
1	Equal importance
2	Weak or slight
3	Moderate importance
4	Moderate plus
5	Strong importance
6	Strong plus
7	Very strong or demonstrated importance
8	Very very strong
9	Extreme importance

Priorities

These are the resulting weights for the criteria based on your pairwise comparisons:

Cat		Priority	Rank	(+)	(-)
1	Elevation	21.8%	1	5.1%	5.1%
2	Slope	20.3%	2	5.7%	5.7%
3	Distance_River	15.8%	3	5.2%	5.2%
4	Rainfall	13.8%	4	5.0%	5.0%
5	Drainage_Density	8.3%	5	2.6%	2.6%
6	Stream Power Index	5.3%	6	1.4%	1.4%
7	LULC	4.7%	7	1.4%	1.4%
8	Ruggedness Index	3.4%	8	1.0%	1.0%
9	Lithology	2.7%	9	1.0%	1.0%
10	TWI	2.2%	10	0.7%	0.7%
11	NDVI	1.7%	11	0.8%	0.8%

Number of comparisons = 55
Consistency Ratio CR = 3.4%

Decision Matrix

The resulting weights are based on the principal eigenvector of the decision matrix:

	1	2	3	4	5	6	7	8	9	10	11
1	1	1.00	2.00	2.00	3.00	5.00	6.00	6.00	7.00	8.00	8.00
2	1.00	1	2.00	2.00	3.00	4.00	5.00	6.00	7.00	6.00	6.00
3	0.50	0.50	1	2.00	3.00	3.00	4.00	5.00	6.00	6.00	7.00
4	0.50	0.50	0.50	1	3.00	4.00	4.00	5.00	5.00	5.00	6.00
5	0.33	0.33	0.33	0.33	1	2.00	3.00	3.00	4.00	4.00	5.00
6	0.20	0.25	0.33	0.25	0.50	1	1.00	2.00	3.00	3.00	4.00
7	0.17	0.20	0.25	0.25	0.33	1.00	1	2.00	2.00	3.00	4.00
8	0.17	0.17	0.20	0.20	0.33	0.50	0.50	1	2.00	2.00	3.00
9	0.14	0.14	0.17	0.20	0.25	0.33	0.50	0.50	1	2.00	3.00
10	0.12	0.17	0.17	0.20	0.25	0.33	0.33	0.50	0.50	1	2.00
11	0.12	0.17	0.14	0.17	0.20	0.25	0.25	0.33	0.33	0.50	1

Principal eigen value = 11.521
Eigenvector solution: 5 iterations, delta = 3.0E-8

Figure 14 Priority and decision matrix for all the parameters.

After creating the matrix, the following Equation (7) has been used to find the maximum eigenvalue (λmax):

$$\lambda = \frac{E}{K}. \quad (7)$$

Where:

E = priority vector,
K = number of parameters

Determine the consistency index (CI) using the Equation (8) below:

CI = (λmax − K)/(K − 1) = (11.521 − 11)/10 = 0.0521

Where:

λmax = Principal eigenvalue,

K =No. of parameters,

(λmax) = 11.521, and

CI = (11.521/10) = 0.0521.

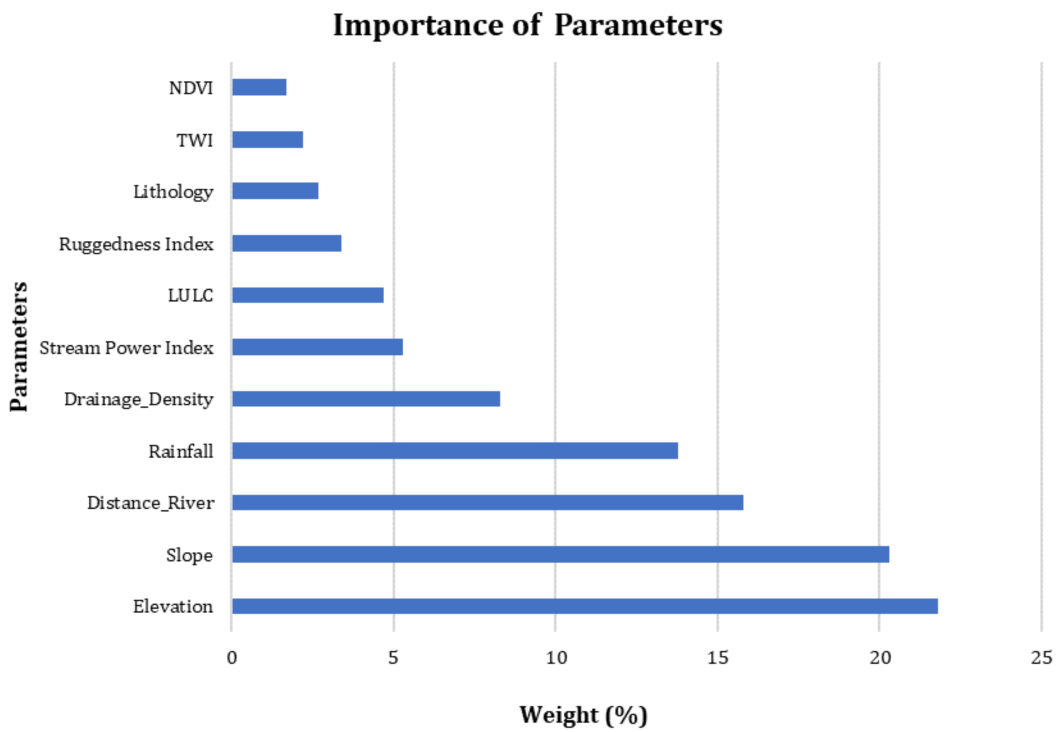


Figure 15 Weights (%) of all the parameters.

Figure 15. represents the importance and weight in (%) of each parameter separately. The map for each layer has been generated and classified. Elevation exerts the most significant influence on flooding, accounting for 21.8%; slope follows as the second most impactful factor at 20.3%; DFR ranks third with 15.8%; rainfall is fourth at 13.8%; DD is fifth with 8.3%; SPI ranks sixth at 5.3%; LULC is seventh with 4.7%; ruggedness index is eighth at 3.4%; lithology occupies the ninth position with 2.7%; TWI is tenth at 2.2%; and NDVI has the least impact at 1.7%. The principal eigenvalue (λmax) was determined to be 11.521. The CR value is 3.4 percent.

4.4 Consistency check

According to Saaty (1980), the value of the Consistency Ratio (CR) indicates that the Analytic Hierarchy Process (AHP) does not necessitate transitive or consistent judgements. If the consistency ratio is below 10%, the judgement is deemed consistent; if it surpasses 10%, the evaluations may require revision. Equation (9) is utilized to compute CR:

$$CR = \frac{CI}{RI}.. \quad (9)$$

Where:

(RI) = a random index,

CR= 0.521/1.51= 0.0345.

CI = (λmax – K)/(K – 1) = (11.521 – 11)/10 = 0.0521.

The developed pairwise comparison matrix and its weightings were assessed using the Consistency Ratio (CR), ensuring a value below the recommended threshold of 0.1 (Saaty, 1980). The confidence intervals (CI) and confidence ratios (CR) are calculated as 0.0521 and 0.0345, respectively, after determining the weights for each parameter. The established pairwise comparison matrix and its weightings are considered satisfactory as the consistency ratio (CR) is below the specified threshold of 0.1. The regions identified as most susceptible to flooding were those with the highest composite index scores (Kader et al. 2024).

4.5 Creation of flood susceptibility map

After assigning weights and rankings to each theme layer and class of thematic layers, respectively, the flood-vulnerable map was produced using ArcGIS's raster calculator tool and Equation (10):

$$FVZ = \sum_{i=1}^{n}(W_i * R_i) = (E_i * E_r) + (SL_i * SL_r) + (DR_i * DR_r) + (LT_i * LT_r) + (RF_i * RF_r) + (LULC_i \cdot LULC_r) + (DD_i * DD_r) + (TWI_i * TWI_r) + (RI_i * RI_r) + (SPI_i * SPI_r) + (NDVI_i * NDVI_r) \quad (10)$$

Where:

FVZ = flood vulnerable zones,

E = elevation,

SL =slope,

DR = distance from river,

LT =lithology,

RF =rainfall,

LULC = land use and land cover,

DD = drainage density,

TWI = topographic wetness index,

RI = ruggedness index,

SPI = stream power index

NDVI =. Also

W = weight of a parameter, and

R = rank of classes within each parameter.

4.6 Creation of settlement vulnerability map

To create the flood inundation layer for Khagaria district, dates of flood impact were initially determined using newspaper reports and other reliable sources. Subsequently, Sentinel-1 SAR data with VV and VH polarisations for the specified dates of 02 July 2024, 07 August 2024, 19 August 2024, 31 August 2024, 12 September 2024, 24 September 2024, and 06 October 2024 were obtained. An analysis of pre- and post-event images was performed for each date to define the extents of flood inundation. Date-specific inundation layers were created and subsequently amalgamated to form a comprehensive flood inundation layer for the year 2024. The validation of the layer was conducted using 57 flooded points and 43 non-flooded points. In 2024, settlement areas affected by flooding were examined at the village level, encompassing the extent (in hectares), percentage of impacted settlement area, and frequency of flood events.

4.7 Validation through ROC and AUC

It is essential to verify the accuracy of the data and the outcomes of the flood threat assessments to substantiate the study's findings (Tehrany et al. 2019). Pham et al. (2017) describe the Receiver Operating Characteristic (ROC) curve, which illustrates specificity on the x-axis and sensitivity on the y-axis. The validation process concluded by juxtaposing the generated vulnerability map with actual flood occurrences through the ROC curve and AUC methodologies (Jensen 2004). Quality metrics were derived from the cross-tabulation of referenced and categorised data, known as the error matrix (Congalton & Green 2008). The ROC-AUC was determined by juxtaposing the flood susceptibility map with actual flood and non-flood sites. Flood and non-flood points were gathered from ground surveys and analysed using the Arc-SDM toolbox in the ArcGIS environment.

5 Results and Discussion

5.1 Flood susceptibility analysis

The research employed 11 parameters to generate the final flood map utilizing the AHP technique. The flood susceptibility assessed via the AHP technique is categorised into five classes: very low, low, moderate, high, and very high, each represented by distinct colours as illustrated in Figure 16. Khagaria district encompasses a total area of 1,490.16 square kilometres. The region designated as a very low flood susceptibility zone encompasses 75.362 sq. km, the low flood susceptibility zone covers 212.323 sq. km, the moderate flood susceptibility zone spans 268.844 sq. km, the high flood susceptibility zone measures 429.909 sq. km, and the very high flood susceptibility zone extends over 399.502 sq. km, as illustrated in Table 4.

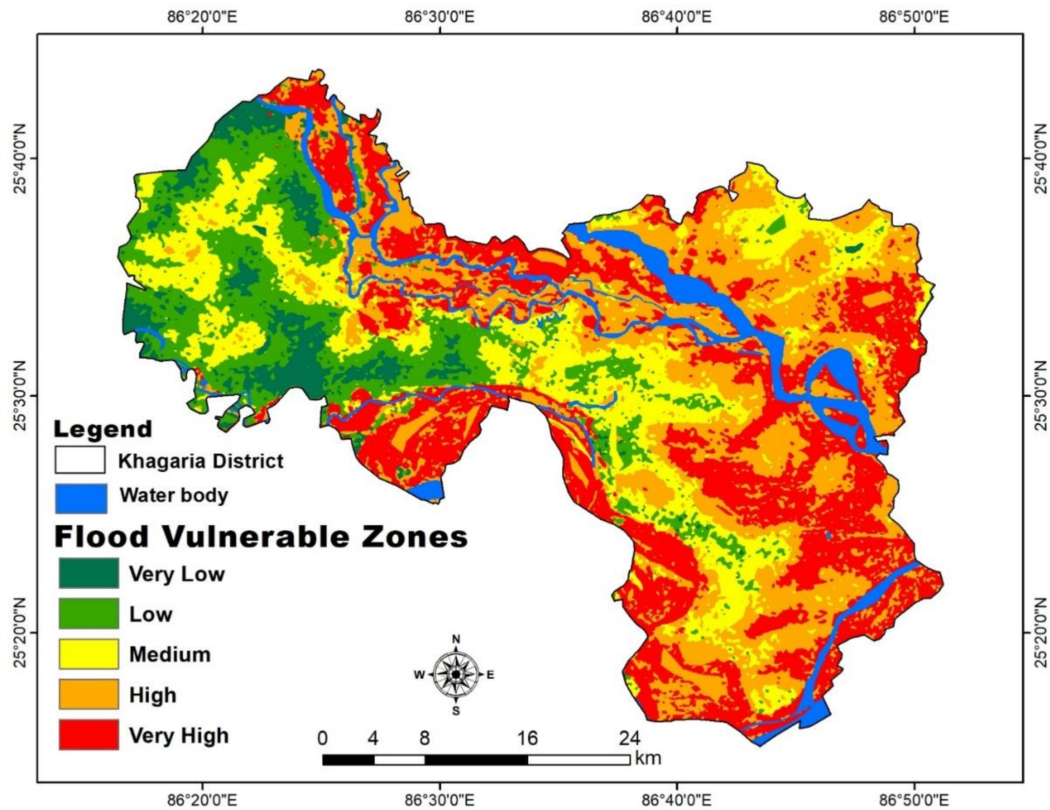


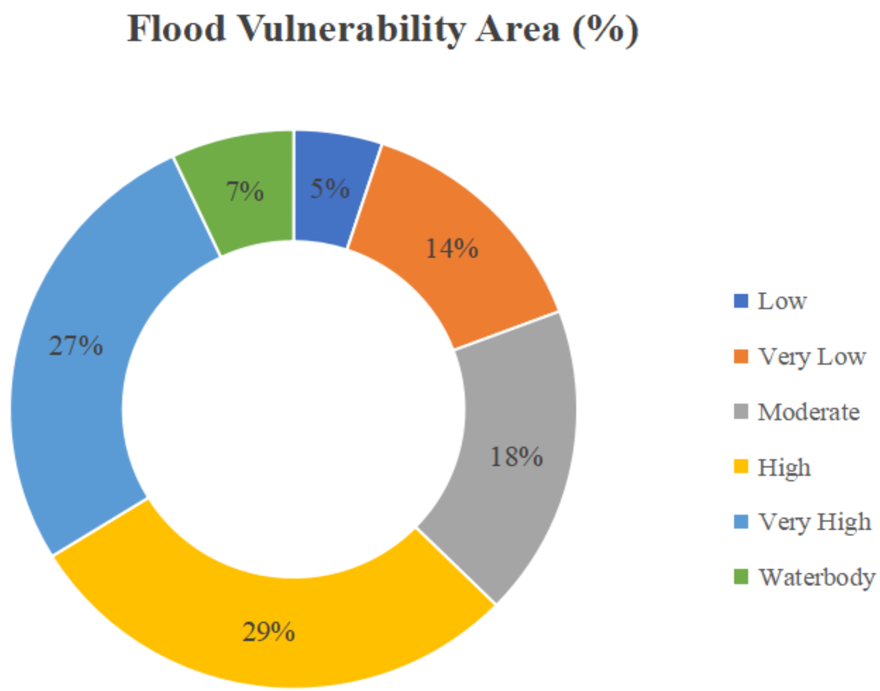
Figure 16 Flood susceptibility map.

Table 4 Flood vulnerable areas with their percentage.

Sr. No	Class	Area (km^2)
	Very Low	75.362
	Low	212.323
	Moderate	268.844
	High	429.909
	Very High	399.502
	Waterbody	104.221
	Total	1490.16

Figure 17 above illustrates the spatial distribution of flood risk zones in Khagaria District, Bihar. The district is categorised into six groups based on flood susceptibility: Very High (27%), High (29%), Moderate (18%), Very Low (14%), Low (5%), and Waterbody (7%). More than half of the region is highly susceptible to flooding, with 56% of the district classified within the High and Very High sensitivity categories. Typically located near low-lying floodplains and riverbanks, these areas often experience flooding during monsoon seasons. Regions with moderate vulnerability (18%) occasionally experience flooding but are typically in a more favourable condition.

Figure 17 Flood susceptibility Area (%).



Conversely, Very Low (14%) and Low (5%) vulnerability zones indicate regions that are less vulnerable to flooding, possibly because of elevation or distance from flood sources. Permanent rivers and wetlands are represented by waterbodies (7%) and are also important in flood dynamics.

Very high flood susceptibility zones

The villages such as Imadpur, Dariapur, Siswa, Siadatpur Aguani Inglish, Siromanharpur, Shirinia, Duraja, Marachi, Katghara, Chandwa, Ragdhan Karari, Rampur, Durgapur, and Sathman are classified as being in the Very High flood susceptibility category. These locations are highly susceptible to recurrent and severe flooding due to their predominance in low-lying areas adjacent to rivers. Their susceptibility to flooding is heightened by proximity to these rivers, level topography, and inadequate drainage.

High flood susceptibility zones

The villages such as Mora, Temtha, Dumaria Bazurg, Madhopur, Zorawarpur, Kharowa, Kulsara, Rohinwa, Araria, Pachpirhia, Ramnagar, Goas, Gobindpur, Rupli Inglish, and Khajraitha are classified as highly vulnerable to flooding. These sites are situated in floodplains or adjacent to riverbanks, where hydrological and topographical characteristics heighten the risk of flooding. Their susceptibility is exacerbated by factors such as land use patterns, soil permeability, and insufficient flood control infrastructure.

Medium flood susceptibility zones

The villages such as Giddha, Dighni, Telaunchh, Bheria, Deori, Kaunia, Balha, Bakhtiarpur, Imli, Dighni, Khutia, Buchai, Bhagwan Chak, Dhatauli, and Chhilkauri exhibit medium flood susceptibility. Moderate flooding may occur in these areas, and it is frequently impacted by local drainage systems and seasonal rainfall patterns. A balanced risk profile is produced by the combination of moderate elevation and partial exposure to flooding.

Low flood susceptibility zones

The villages such as Chainpur Khurd, Rahima, Burhwa, Ranko, Nista, Adabari, Raun, Chharapatti, Gaura Chak, Ahuniswa, Summha, Barai, Tetrabad, Bhadas, Amni, are classified as having low flood susceptibility. These settlements are less vulnerable to floods since they are usually situated at higher altitudes or have efficient flood control systems in place. Their reduced danger levels are a result of better drainage systems and the existence of natural barriers.

Very low flood susceptibility zones

The villages such as Sano Khar, Darhi, Sonhauli, Mathurapur, Seikhpura, Bishunpur Dhusmari, Rani Shakarpura, Alauli, Sobhni, Patti Chitarsari, Mehdipur Labhgaon, Olapur, and Meghauna are classified within the Very Low flood susceptibility category. These locations frequently possess topographical advantages that mitigate the risk of flooding and are situated at a considerable distance from substantial water bodies. They are inherently safeguarded from flooding due to their elevated topography and considerable distance from flood-prone rivers.

Flood inundation mapping for the year 2024 in Khagaria district was conducted utilizing Sentinel-1 SAR data (VV and VH polarisations). Images from before and after the flood were analyzed to identify inundated regions, and all flood event rasters were amalgamated to produce a composite inundation layer. This layer was superimposed on the settlement layer to measure the flood-impacted settlement area at the village level in hectares (Ha). Villages were classified according to the extent of inundated settlement area: ≤5 Ha (26 villages), 5.1–8 Ha (14 villages), 8.1–14 Ha (12 villages), 14.1–22 Ha (6 villages), and >22 Ha (2 villages), as illustrated in Figure 18.

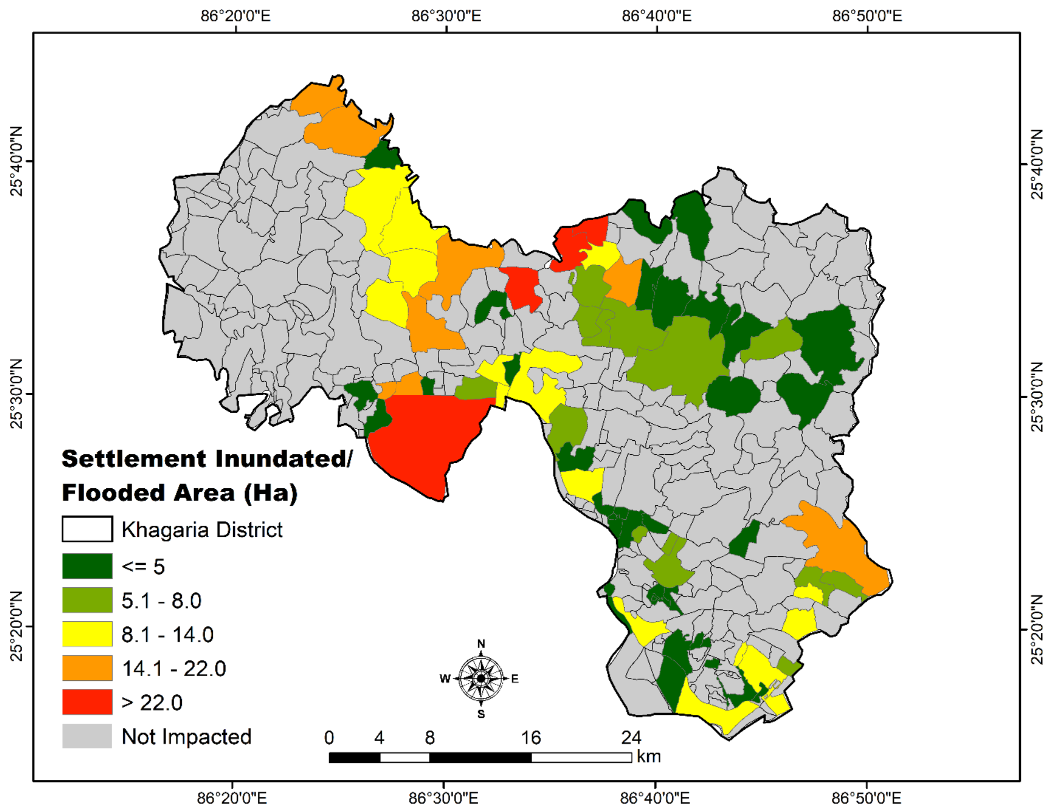


Figure 18 Settlement Inundated/ Flooded Area (Ha) in 2024.

This map illustrates the inundated flood layer of Khagaria, depicting the percentage of settlement affected by flooding across six categories. The dark green colour signifies greater than 14 percent, light green indicates 14.1-31 percent, yellow represents 31.1-54 percent, orange denotes 54.1-80 percent, red reflects above 80 percent, and grey indicates unaffected villages. Nineteen villages have less than 14% of their settlement area affected by flooding, eleven villages have 14.1-31% affected, ten villages have 31.1-54% affected, twelve villages have 54.1-80% affected, and eight villages have 80% or more of their settlement area affected by flooding, as illustrated in Figure 19.

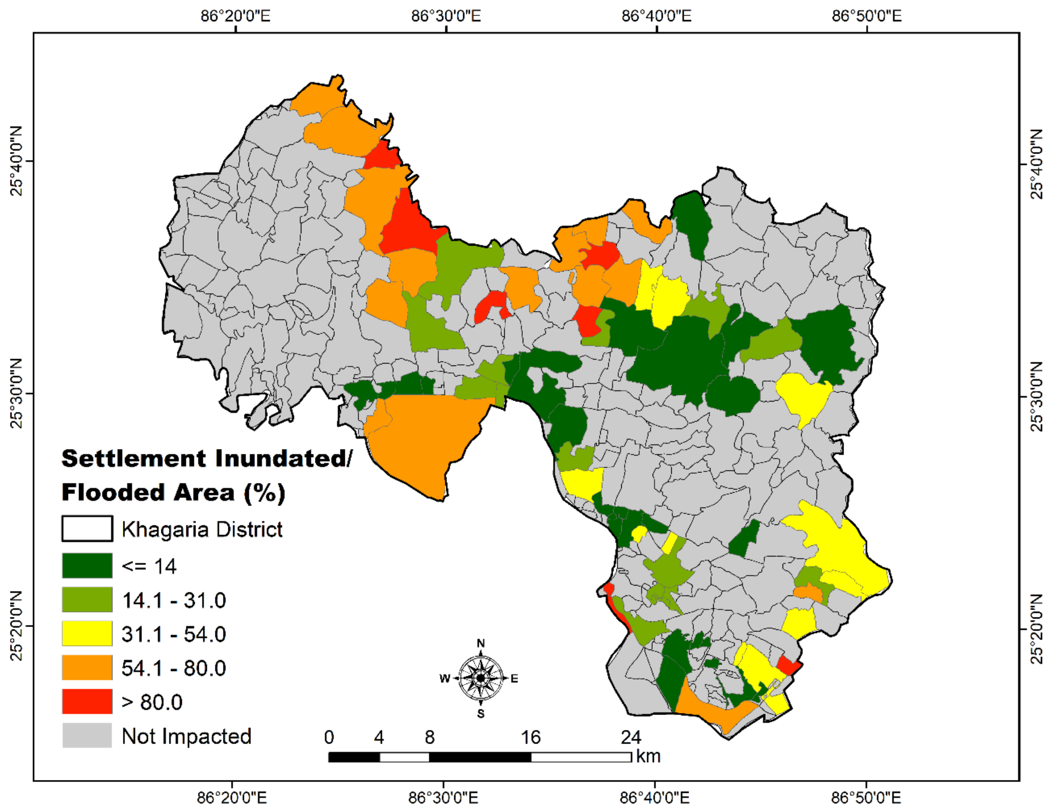


Figure 19 Settlement inundated/ flooded area (%) in 2024.

The subsequent layer was developed utilizing settlement data and an integrated inundation layer generated from Sentinel-1 pre- and post-event data for the entire year of 2024. The village boundary was superimposed on this map to assess the magnitude of the total settlement area inundated by the village in 2024. The red layer indicates settlement in Khagaria District, while the blue layer represents the inundation integrated layer for 2024, as illustrated in Figure 20.

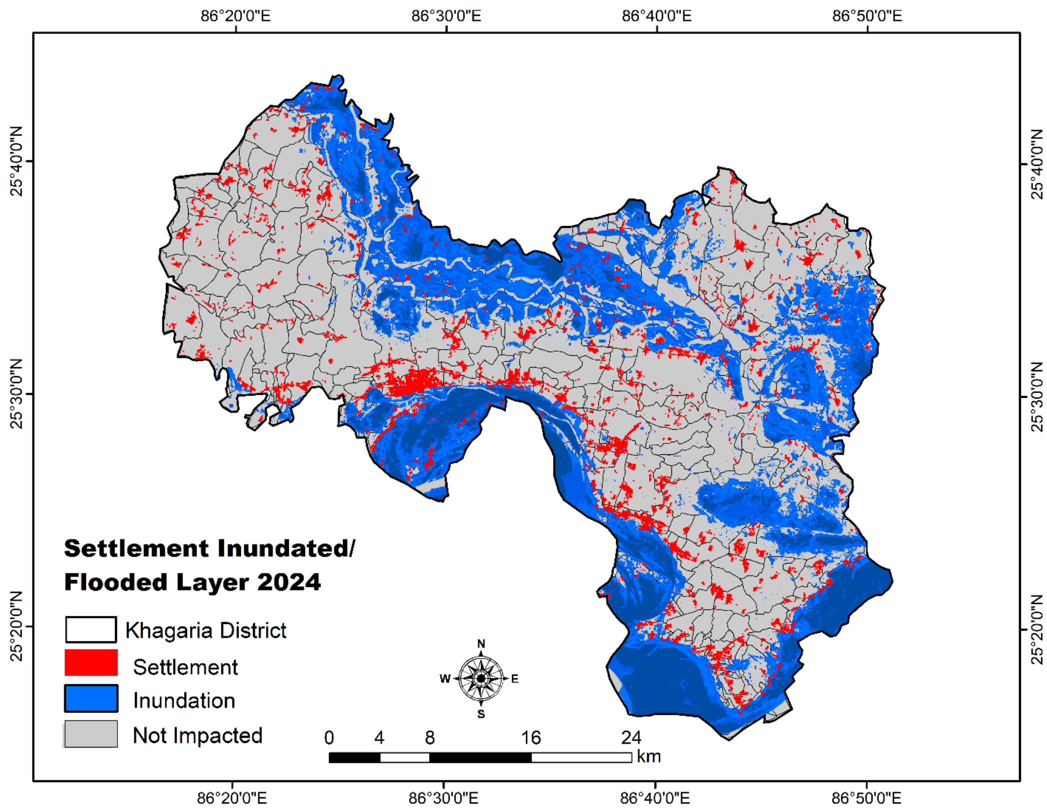


Figure 20 Settlement and Inundated/ Flooded Layer year 2024.

The subsequent map layer illustrates the frequency of settlement inundation/flooding categorised into five groups: <= 2 times (21 villages), 3 times affected (12 villages), 4 times affected (10 villages), 5 times affected (9 villages), and 6 or more times affected (8 villages), as depicted in Figure 21.

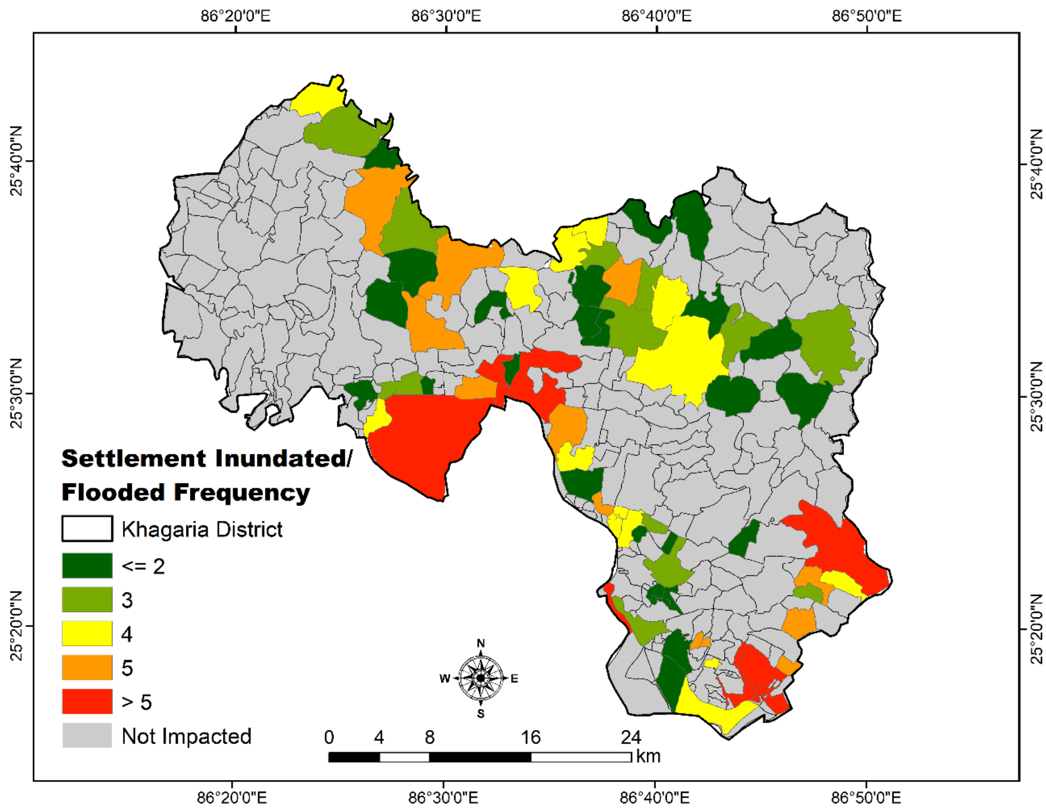


Figure 21 Settlement and inundated/ frequency map in 2024.

This research indicates that Khagaria is among the most flood-prone regions, as a significant portion of the study area falls within a very high and sensitive zone classification. This classification demonstrates the significance of incorporating various thematic layers and hydrological studies in flood risk analysis. To safeguard individuals, agriculture, and infrastructure in this flood-prone region, Khagaria urgently requires flood mitigation strategies, encompassing embankments, disaster preparedness, and land-use management, as indicated by this vulnerability mapping. To alleviate the effects of severe flooding in at-risk areas, targeted flood control strategies, including the enhancement of drainage systems and improved land use regulation, must be implemented. The data supports the necessity for prompt intervention: afforestation, flood zoning, robust infrastructure, awareness campaigns, and additional strategies. Moreover, the geographic distribution aids government agencies, NGOs, and researchers in formulating targeted strategies for disaster preparedness, sustainable land use planning, and climate change adaptation in this vulnerable region. The study's findings support the integration of AHP and GIS approaches, which provide a potent tool for flood risk mapping decision-making processes since these techniques enable a logical and effective use of the spatial data.

Based on the results it is recommended to make people make aware of low-lying areas near rivers and make the, stay away from those low-lying areas. There is a need of strong infrastructure in the districts such as building embankments, dams, houses for poor in low lying areas.

Table 5. comparison between the AHP-derived flood vulnerability classes and frequently inundated villages and settlements

Sr. No.	Flood Vulnerability Class (AHP Map)	Number of Frequently Inundated Villages	Number of Frequently Inundated Settlements
1	Very Low	1	1
2	Low	5	3
3	Moderate	9	11
4	High	29	36
5	Very High	31	24
	Total	**75**	**75**

The integrated comparison between the AHP-derived flood vulnerability classes and frequently inundated villages and settlements shows a clear and consistent spatial correspondence between modeled vulnerability and observed flood occurrence. A dominant concentration of inundation is observed in the higher vulnerability classes, particularly “High” and “Very High,” which together account for 60 villages and 60 settlements out of the total 75, indicating that most flood-affected units are concentrated in high-risk zones identified by the AHP framework.

At the village level, 29 villages fall under the “High” class and 31 under the “Very High” class, while at the settlement level, 36 settlements and 24 settlements respectively fall within these same categories. This confirms that the majority of frequently inundated settlements are spatially aligned with areas classified as highly vulnerable, demonstrating strong agreement between the AHP-based vulnerability zoning and actual flood impacts. The relatively small representation of lower vulnerability classes, where “Very Low” and “Low” together account for only 6 villages and 4 settlements, further strengthens this pattern by indicating minimal inundation occurrence in low-risk zones. Results confirm a strong and systematic relationship between increasing AHP vulnerability class and Actual flood inundation Frequency 2024, thereby validating the robustness and practical reliability of the AHP-based flood vulnerability assessment.

5.2 Validation through ROC and AUC method

A total of 250 random points were selected for validation, comprising 150 points in flooded areas and 100 points in non-flooded areas, collected via ground survey using mobile GPS. Receiver operating characteristic (ROC) curve analysis was conducted using the Arc SDM module in ArcGIS software to evaluate the accuracy of the vulnerability map's predictions. Figure 22 illustrates the flooded and non-flooded points, while Figure 23 presents the graph for AUC and ROC.

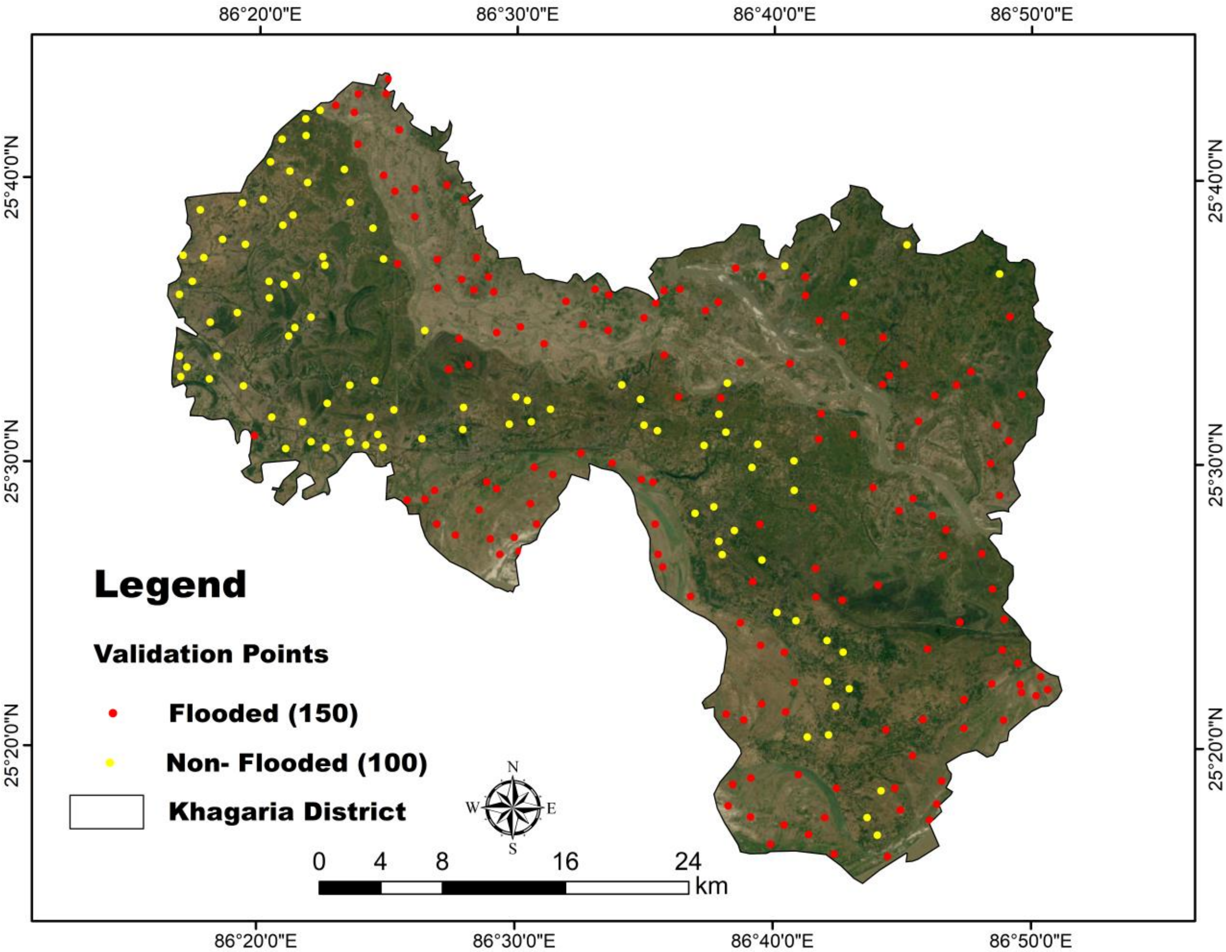


Figure 22 Flood Validation points map.

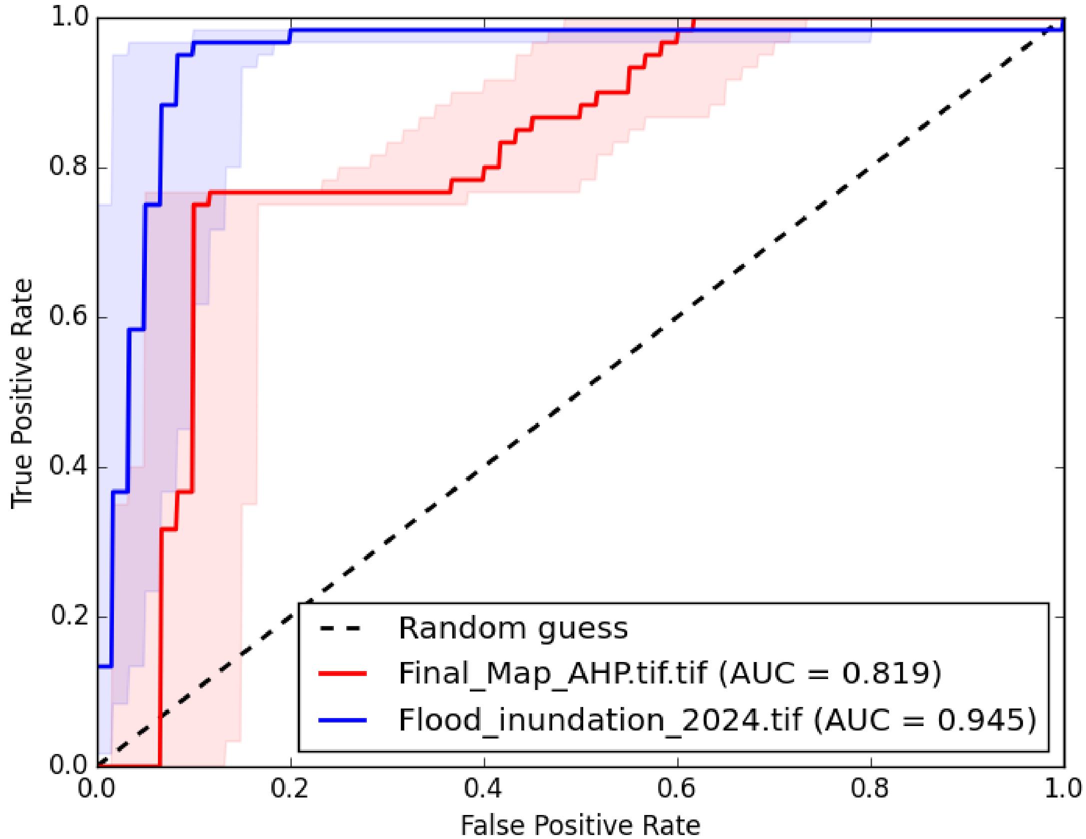


Figure 23 ROC and AUC curve.

Random chance is represented by an area under the ROC curve (AUC) of 0.5, while perfect precision is indicated by an AUC of 1.0 (Zou et al. 2007). The AUC values for predictive accuracy are as follows, as reported by Andualem & Demeke (2019): Excellent (0.90–1.00), Very Good (0.80–0.90), Good (0.70–0.80), Average (0.60–0.70), and Low (0.50–0.60). The AUC, ranging from 0 to 1, depicts the correlation between sensitivity and specificity. A higher AUC score signifies superior model quality, while a lower score indicates poor performance (Shingute et al. 2024). The importance of the Area Under the Curve (AUC) was evaluated using Receiver Operating Characteristic (ROC) analysis. The flood inundation layer attained an AUC value of 0.945, whereas the flood susceptibility zonation map registered an AUC of 0.819. These values demonstrate robust predictive performance and a significant correlation between satellite-derived flood data and the identified flood-prone areas.

6 Conclusion

The study focuses on the Khagaria district, a highly flood-prone region in Bihar, which has been largely overlooked in prior geospatial flood research, thereby addressing a significant knowledge deficiency. The objective is to identify flood-prone areas in Khagaria district by utilizing a multicriteria decision-making approach within a geographical information system, incorporating remotely sensed data. The final flood map was generated using the AHP method, which utilized an 11 × 11 decision matrix to denote the importance of the following parameters: elevation, slope, rainfall, DD, DFR, TWI, SPI, LULC, NDVI, lithology, and roughness index. Khagaria district encompasses an area of 1,490.16 km^2, with 75.362 km^2 designated as very low flood-vulnerable, 212.323 km^2 as low flood-vulnerable, 268.844 km^2 as moderately flood-vulnerable, 429.909 km^2 as highly flood-vulnerable, and 399.502 km^2 as very highly flood-vulnerable. Villages such as Imadpur, Dariapur, Siswa, Siadatpur Aguani Inglish, Siromanharpur, Shirinia, Duraja, Marachi, Katghara, Chandwa, Ragdhan Karari, Rampur, Durgapur, and Sathman are categorised as exhibiting extreme flood v susceptibility due to their geographical area. The integrated 2024 flood inundation raster was utilized to determine the village-specific affected settlement areas (in hectares and percentage) and their inundation frequency, resulting in the preparation of maps for the year 2024. The area under the curve (AUC) for the inundation layer is 0.945, while the AUC for the flood susceptibility zonation map is 0.819 which is acceptable. The results can substantially enhance local flood risk mitigation strategies, infrastructure advancement, land-use planning, and emergency preparedness initiatives.

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